

1. a) Install Flutter and Dart SDK

Flutter Installation in Windows

System Requirements:

Operating System: Windows 10 or Later (You can also use Mac or Linux OS.).

Disk Space: 400 MB (It does not include disk space for IDE/tools).

Windows PowerShell

Git for Windows

SDK: Flutter SDK for Windows

IDE: Android Studio (Official)

Install Git:

Step1: To download Git

Step2: Run the .exe file to complete the installation. During installation, make sure that you have selected the recommended option.

Install Flutter SDK:

Step-1: Download the installation bundle of the Flutter Software Development Kit for windows.

To download Flutter SDK, Go to its official website, click on Get started button

Step-2: Next, to download the latest Flutter SDK, click on the Windows icon.

Here, you will find the download link for SDK.

Step-3: When your download is complete, extract the zip file and place it in the desired installation folder or location.

Step-4: To run the Flutter command in regular windows console, you need to update the system path to include the flutter bin directory. The following steps are required to do this:

Step-4.1: Go to My Computer → properties → advanced tab → environment variables.

Step-4.2: Now, select path → click on edit.

Step-4.3: In the above window, click on New → write path of Flutter bin folder in variable value click ok.

Step-5: Now, run the flutter doctor command. This command checks for all the requirements of Flutter app development and displays a report of the status of your Flutter installation.

Step-6: When you run the above command, it will analyse the system and show its report. Here, you will find the details of all missing tools, which required running Flutter as well as the development tools that are available but not connected with the device.

Step-7: Install the Android SDK. If the flutter doctor command does not find the Android SDK tool in your system, then you need first to install the Android Studio IDE. Install by doing following steps:

Step-7.1: Download the latest Android Studio executable or zip file from the official site.

Step-7.2: When the download is complete, open the .exe file and run it.

Step-7.3: Follow the steps of the installation wizard.

Step-7.4: After Installation is complete, click Next → Finish. Once the Finish button is clicked, you need to choose the Don't import Settings option and click OK. It will start the Android Studio.

Step-8: Next, you need to set up an Android emulator. It is responsible for running and testing the Flutter application.

Step-8.1: To set an Android emulator, go to Android Studio → Tools → Android → AVD Manager and select Create Virtual Device. Or, go to Help → Find Action → Type Emulator in the search box

Step-8.2: Choose your device definition and click on Next.

Step-8.3: Select the system image for the latest Android version and click on Next.

Step-8.4: Now, verify the all AVD configuration. If it is correct, click on Finish

Step-8.5: Last, click on the icon pointed into the red colour rectangle.

Step-9: Now, install Flutter and Dart plugin for building Flutter application in Android Studio. These plugins provide a template to create a Flutter application,

give an option to run and debug Flutter application in the Android Studio itself. Do the following steps to install these plugins.

Step-9.1: Open the Android Studio and then go to File → Settings → Plugins.

Step-9.2: Now, search the Flutter plugin. If found, select Flutter plugin and click install . When you click on install, it will ask you to install Dart plugin, click yes to proceed.

Step-9.3: Restart the Android Studio.

b) Write a simple Dart program to understand the language basics

write any on sample dart program.